(a) one command, then every blank says what to compute

```
$ kinstep new --template her her.kinstep.toml
✓ wrote her.kinstep.toml - hydrogen evolution, HER (Volmer-Heyrovsky-Tafel)
   all 3 mechanism steps are filled in; 6 energy slots are left for you (each one…
   · the Tafel slope identifies the rate-determining step: Volmer ~120, Heyrovsky…
$ cat her.kinstep.toml
# her - hydrogen evolution, HER (Volmer-Heyrovsky-Tafel)
# ! This template carries NO energy values - the mechanism is public knowledge, t…
[mechanism]
H+ + e- + * <-> H*            {beta_ct: 0.5}
H* + H+ + e- -> H2 + *        {beta_ct: 0.5}
2H* -> H2 + 2*
[conditions]
T_K = 298.15
# (1) H+ + e- + * <-> H*
#     what to compute: Volmer step - the key quantity is dG_H* (hydrogen adsorpti…
# Ga_fwd = ?   # forward activation free energy (eV) - uncomment once filled in
# dG_rxn = ?   # reaction free energy (eV)
exit code 0
```

(b) structural gates pass, the provenance gate refuses (exit code 3)

```
$ kinstep check her.kinstep.toml
mechanism: 3 steps, 5 species, site type(s) ['default']
  (1) H+ + e- + * <-> H*  [electrochemical, nu_elec=-1]
  (2) H* + H+ + e- -> H2 + *  [electrochemical, nu_elec=-1]
  (3) 2H* -> H2 + 2*
  ✓ all four gates passed (element/charge balance, site balance, reserved-name cl…
  temperature T_K = 298.15 K
✗ provenance gate rejected the input:
[6 value(s) missing]
  step 1 / Ga_fwd -- step 1 uses transition-state theory and needs the forward ac…
  step 1 / dG_rxn -- step 1 needs the reaction free energy dG_rxn (eV) = G(produc…
  (steps 2 and 3 list the same two slots; 6 in total)
exit code 3
```

(c) one sentence in, structure and questions out (rule-based, no LLM)

```
$ kinstep ask "hydrogen evolution at pH 0, -0.1 V vs SHE, 25 C"
  matched system: her —— hydrogen evolution, HER (Volmer-Heyrovsky-Tafel) [electr…
  values extracted from your sentence (all tagged user_given):
     U_elec = -0.1  <- from "-0.1 v"
     U_ref = SHE  <- from "vs SHE" (reference scale (not a number))
     pH_val = 0  <- from "ph 0"
  ⋆ still needed from you (this module will never guess these):
     · the energies of all 3 steps (Ga_fwd and dG_rxn per step, in eV) - this is …
     ·     (1) H+ + e- + * <-> H*   <- Volmer step - the key quantity is dG_H* (h…
     · temperature (K or degC); the typical value for this system is 298.15 K, bu…
     · active-site areal density gamma_site (mol/m2), used to convert the surface…
exit code 0
```

(d) solve, then report coverages, precision limits, DRC and self-checks

```
$ kinstep run ammonia.kinstep.toml --drc
steady state (ode(4 continuation stages up to 4.9e+03 s) → damped-newton(1 steps,…
  coverages:  H*=0.219289  N*=6.07382e-07  NH*=3.81309e-06  NH2*=0.0025178
  vacancies:  *=0.778189
  scaled residual ||F||_inf = 1.855e-50
     step 1  1-z=1.027e-13    attainable rel. precision 9.7e-04   (~3 digits)  <-…
  ✓ flux-conservation self-check: max relative residual 4.09e-17
     step 1  H2 + 2* <-> 2H*              X_B = -0.000000
     step 2  N2 + 2* <-> 2N*              X_B = +0.999998
     step 5  NH2* + H* <-> NH3 + 2*       X_B = +0.000000
     ⋆ sum rule  sum X_B = 1.000000000 (should be 1; deviation 2.9e-13)
  reaction orders: H2: -0.2033  N2: +1.0045  NH3: -0.0140
  apparent activation energy Ea_app = 3.016887 eV
exit code 0
```